

**UNIVERSIDADE POSITIVO
ESCOLA POLITÉCNICA
ENGENHARIA ELÉTRICA
ENGENHARIA DA COMPUTAÇÃO
CIRCUITOS ELÉTRICOS II**

**MEMORIAL DESCRITIVO:
PROJETO INTERRUPTOR POR PALMAS**

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1 INTRODUÇÃO

A realização desse projeto visa atender a atividade do primeiro bimestre da disciplina de Circuitos Elétricos II dos cursos de Engenharia Elétrica e da Computação da Universidade Positivo. A proposta docente é construir um circuito prático com pelo menos dez elementos básicos e a presença de capacitores ou indutores. É necessário atestar o perfeito funcionamento, realizar este Memorial Descritivo e apresentar em aula. Ao realizar as atividades, o aluno se defrontará com os desafios das diversas fases do projeto e as atividades comporão a aprendizagem e treinamento do futuro engenheiro.

2 MEMORIAL DESCRITIVO

Este projeto tem por objetivo inovar e trazer soluções interessantes no chaveamento de lâmpadas ou mecanismos elétricos.

2.1 DEMANDA

A opção de chaveamento por palmas responde a necessidade de acionar algum elemento elétrico sem fazer contato direto com o interruptor. Não importa qual a razão da impossibilidade de se tocar no interruptor.

2.2 DESCRIÇÃO DAS ATIVIDADES

Para executar esse projeto, seguimos o seguinte processo:

- a) Pesquisa de circuitos já disponíveis e definição de qual atenderia melhor ao projeto previamente especificado pelo professor de Circuitos Elétricos II, Jackson Milano;
- b) Verificação em aula com o professor se o circuito escolhido atende aos critérios da proposta;
- c) Teste em matriz de contatos e validação de funcionamento do referido circuito;
- d) Desenvolvimento do *layout* da placa de circuito impresso.
- e) Fabricação na universidade;
- f) Montagem dos componentes e testes finais para assegurar o devido funcionamento;
- g) Elaboração do Memorial Descritivo;

- h) Elaboração do material de apresentação em aula.

2.3 RECURSOS E MATERIAIS UTILIZADOS

O que precisamos para realizar esse projeto é:

2.3.1 Composição do circuito e especificações com base nos *datasheets*:

Destacamos alguns dados principais das especificações dos elementos utilizados no circuito. Na sessão anexo, ao final deste memorial, também trouxemos os *datasheets* na íntegra dos componentes mais relevantes.

- a) Borne modular de 2 vias

Fabricante	Metaltex
Modelo	BR102A
Tensão nominal	250 V
Corrente	16 A
Passo	5 mm
Resistencia de contato	20 mΩ
Resistencia de isolação	500 MΩ / 1000 VDC

O borne faz a conexão entre os fios da lâmpada e o Relé. Quando o Relé é acionado, o circuito é fechado e a lâmpada acende. Custo aproximado de: R\$ 1,50 cada.

- b) Capacitor eletrolítico radial 47 μF/35 V

Fabricante	United Chemi-Con
Serie	LE
Dielétrico	Alumínio
Temperatura	-10 °C a +105 °C
Tolerância	20%
Espaço entre terminais	2,5 mm

Este capacitor suavizará as alterações da passagem da corrente entre os terminais não permitindo variações bruscas. Custo aproximado de: R\$ 0,20 cada.

c) Clip conector de 2 vias para bateria de 9 V

Fará a conexão da bateria com o circuito para fornecer a energia. Custo aproximado de: R\$ 2,50 cada.

d) Circuito integrado 555

Fabricante	Texas Instruments
Modelo	LM555 <i>Timer</i>
Tensão	5 a 15 V
Corrente	15 mA (máximo)

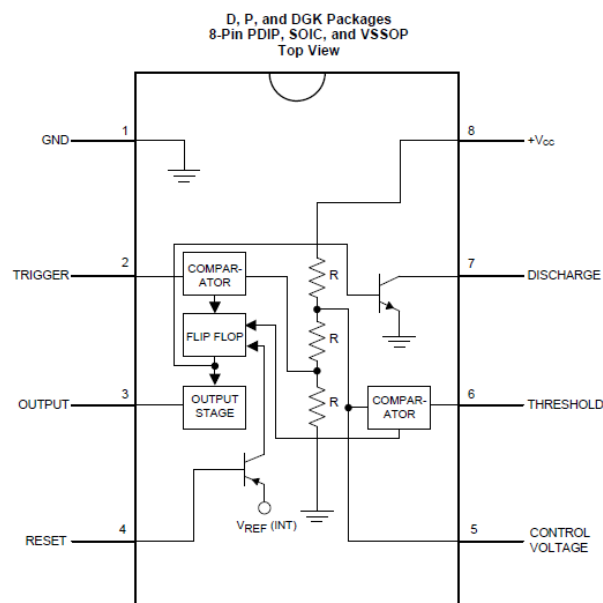


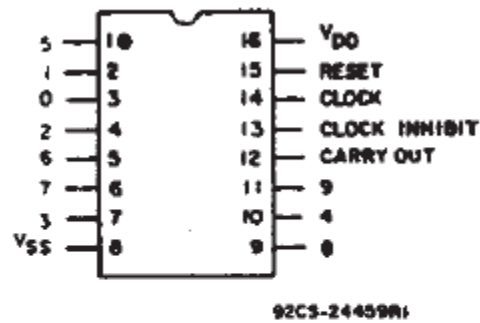
FIGURA 1 – PINAGEM CI 555
FONTE: TEXAS INSTRUMENTS, 2015

O LM555 é um dispositivo altamente estável para gerar oscilações ou atrasos de tempo precisos. Seus terminais estão dispostos como nos mostra a Figura 1 e trazem a opção de acionar (trigger) ou redefinir (reset), conforme o desejado. No circuito, ele fará a oscilação do sinal para o contador CD4017. Custo aproximado de: R\$ 1,50 cada.

e) Circuito integrado CD4017

Fabricante	<i>Texas Instruments</i>
------------	--------------------------

Modelo	CD4017 Counter
Tensão	5 a 15 V
Frequência Clock	2,5 a 5,5 MHz
Pulso	60 a 200 ns



**TOP VIEW
CD4017B
TERMINAL DIAGRAM**

FIGURA 2 – PINAGEM CI CD4017
FONTE: TEXAS INSTRUMENTS, 2004

O contador CD4017 é um contador de década. Seus terminais, como mostra a Figura 2, fornecem sinais para os 9 dígitos (0 – 9). A depender do nível do sinal do oscilador LM555, ele emitirá ao Relé o sinal que acenderá ou apagará a lâmpada. Custo aproximado de: R\$ 1,50 cada.

f) Diodo retificador 1N4004

Fabricante	Vishay
Tensão reversa max. de pico	400 V
Tensão máx. eficaz	280 V
Tensão DC máx. de bloqueio	400 V
Retificação máx. de Corrente	1 A
Pico de corrente (onda senoidal)	30 A
Pico de corrente (onda quadrada)	45 A

Uma das aplicações do diodo é na retificação de corrente conduzindo-a para apenas um sentido e bloqueando o caminho reverso. Evitando danificar outros componentes. É esse intuito para o circuito. Custo aproximado de: R\$ 0,15 cada.

g) LED verde

Fabricante	Symtronic
Modelo	SYM-G503-30-D
Potência dissipada	60 mW
Corrente de pico	80 mA
Corrente de operação	15 a 20 mA

Utilizado nesse projeto para representar seu funcionamento. Custo aproximado de: R\$ 0,25 cada.

h) Microfone de Eletreto com 2 terminais

Sensibilidade	48 a 66 dB
Tensão de entrada	3 V
Tensão de operação	1,5 a 10 V
Consumo de corrente	0,5 mA
Pressão sonora	120 dB

Esse microfone é a porta de entrada do sinal da palma para acionamento do circuito. Custo aproximado de: R\$ 1,50 cada.

i) Bateria alcalina de dióxido de manganês 9 V

Tensão nominal	9 V
Impedância	1,7 mΩ
Terminais	Estalo miniatura

A bateria de 9 V é responsável por alimentar o circuito. Com exceção da lâmpada que é alimentada pela tensão da casa (tomada). Custo aproximado de: R\$ 5,00 cada.

j) Placa de Fenolite cobreada face simples 10 cm de largura por 10 cm de comprimento

A placa é a base onde construímos o circuito e soldamos os componentes. Custo aproximado de: R\$ 5,00 cada.

k) Relé 12 V

Fabricante	Metaltex
Modelo	AZ1RC2
Tensão nominal	12 V
Tensão de operação	acima de 9 V
Tensão de não operação	abaixo de 1,2 V
Corrente nominal	30 mA
Resistência	400 Ω

O Relé faz a conexão do circuito com a lâmpada mantendo as tensões do circuito com a da casa independentes. Quando recebe o sinal ele fecha a chave e aciona a lâmpada. Custo aproximado de: R\$ 4,00 cada.

l) Resistores:

Filme carbono: 1 k Ω ; 10 k Ω ; 100 k Ω

Serie	E24
Tolerância	5%
Dissipação nominal	0,5 W
Resistencia térmica	170 K/W
Tensão de operação máx.	200 V

Os resistores fazem o ajuste da corrente para o correto funcionamento e proteção dos componentes com ele conectados. Custo aproximado de: R\$ 0,15 cada.

Potenciômetro: 100 k Ω

Fabricante	Alpha
Serie	31VA
Tolerância	20%
Tensão máxima	500 V
Dissipação	0,5 W

Com o potenciômetro regulamos a intensidade da captação sonora pelo microfone. Custo aproximado de: R\$ 1,70 cada.

m) Soquete estampado para circuitos integrados

Neste projeto, utilizamos esses soquetes para encaixar os circuitos integrados LM555 e CD4017. Eles auxiliam na manutenção por deixar os componentes encaixados e sem a necessidade de soldar. Assim, no eventual defeito, basta retirar o componente encaixado. Custo aproximado para o de 8 pinos: R\$ 0,30 cada. Custo aproximado para o de 16 pinos: R\$ 0,40 cada.

n) Transistor BC548

Fabricante	Fairchild
Tensão coletor - base	50 V
Tensão coletor – emissor	30 V
Tensão emissor – base	5 V
Corrente coletor	100 mA
Dissipação coletor	500 mW

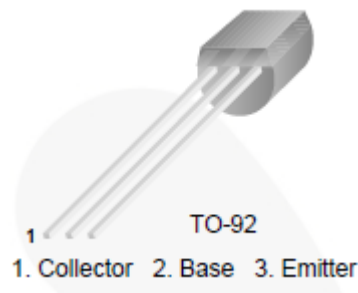


FIGURA 3 – DESENHO E TERMINAIS DO TRANSISTOR
FONTE: FAIRCHILD, 2014

Um transistor amplifica o sinal do microfone e envia para o LM555. O outro recebe o sinal da saída do CD4017 e amplifica a intensidade para acionar o Relé. Pela Figura 3, visualizamos os terminais deste componente. Custo aproximado de: R\$ 0,30 cada.

2.3.2 Confeção da placa de circuito impresso:

A fabricação da PCI, após terminado o *layout*, foi realizada na Universidade Positivo. Para isso, é necessário atender alguns requisitos conforme exposto no item 2.5 deste memorial. O valor para esse serviço foi de R\$ 10,00.

2.3.3 Equipamentos e ferramentas:

- a) Alicates de bico e de corte
- b) Estanho para solda
- c) Ferro para soldar
- d) Fonte de tensão CC ajustável de 0 a 32 V (para testes)
- e) *Jumper*
- f) Matriz de contatos
- g) Multímetro digital

2.4 DIAGRAMA ELÉTRICO DO CIRCUITO

As conexões entre os componentes estão dispostas conforme demonstra a Figura 4 abaixo:

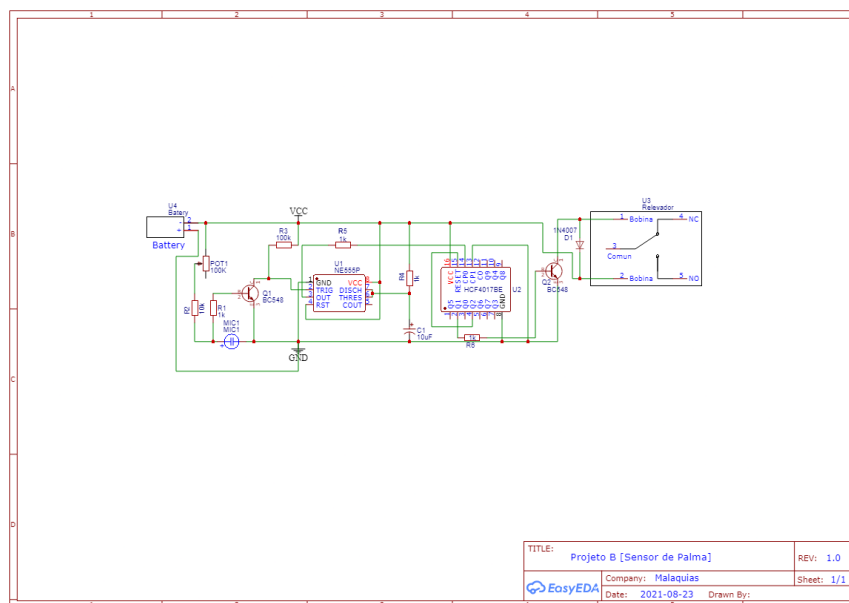


FIGURA 4 – ESQUEMA ELÉTRICO DO PROJETO
FONTE: OS AUTORES

2.5 LAYOUT DA PLACA DE CIRCUITO IMPRESSO

O layout final do circuito está demonstrado na Figura 5 abaixo:

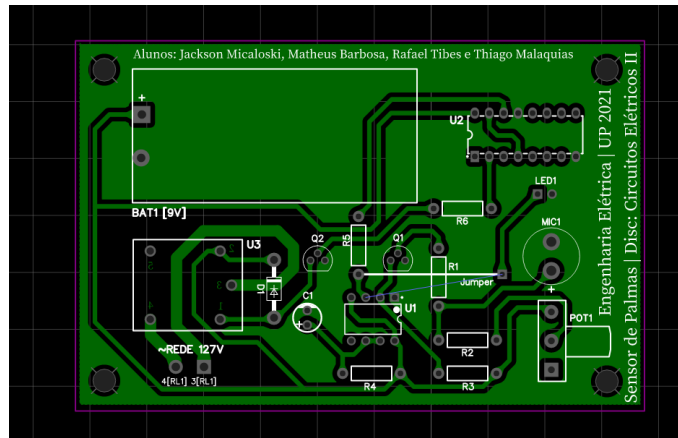


FIGURA 5 – LAYOUT DO PROJETO
FONTE: OS AUTORES

Para que fosse possível fabricar a PCI na Universidade, desenvolvemos, a pedido do fabricante, o *layout* no *software EasyEDA* e geramos o arquivo em formato *Gerber*. O tamanho máximo da utilização da placa não excede a 10 cm de largura por 10 cm de comprimento. O circuito todo foi preparado numa única face. Os furos, para os terminais dos componentes, são de 1 mm de diâmetro e as trilhas de 0,8 mm de espessura.

A Figura 6 é o resultado da placa com as trilhas elétricas e as furações para os terminais dos componentes.

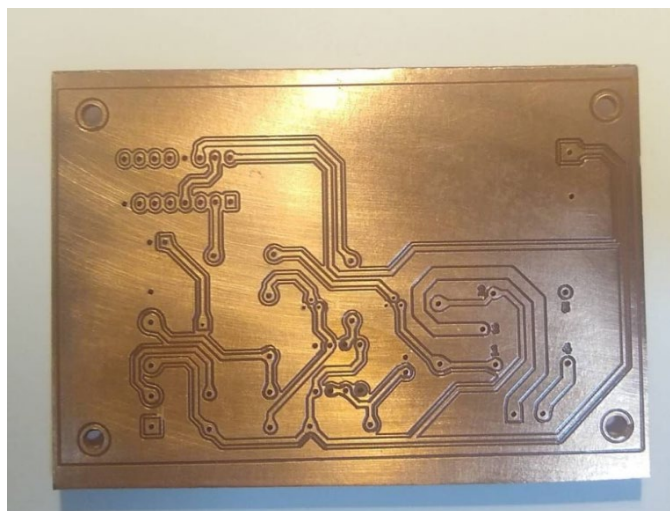


FIGURA 6 – PRODUÇÃO DA PCI
FONTE: OS AUTORES

2.6 MONTAGEM

A montagem dos componentes na placa está representado pela Figura 7. Nessa fase, posicionamos cada componente, em seu devido lugar, e soldamos na placa. Tomando os devidos cuidados com o ferro de solda, por sua elevada temperatura de trabalho, e com a fumaça tóxica gerada pelo estanho.



FIGURA 7 – MONTAGEM FINAL DO PROJETO
FONTE: OS AUTORES

Um dos integrantes do projeto, teve a oportunidade de projetar e imprimir, em uma impressora 3D, através do *software SolidWorks (2017)*, um “case” para a placa eletrônica. A Figura 8 demonstra como ficou a placa alocada neste “case”.

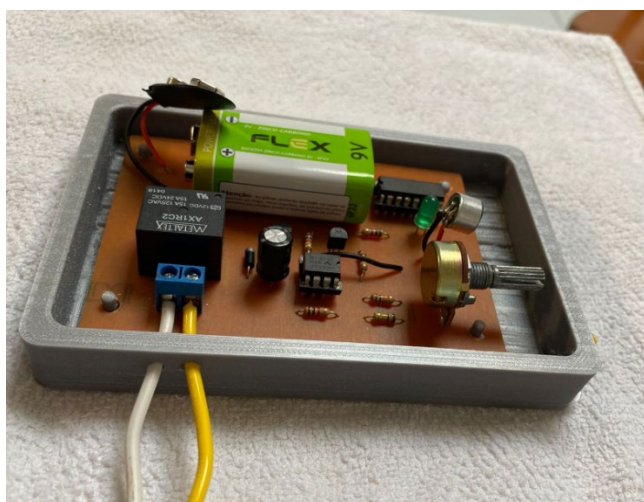


FIGURA 8 – CASE DO PROJETO
FONTE: OS AUTORES

2.7 CUSTOS

A Tabela 1 a seguir é um panorama geral dos custos com os componentes necessários para a execução do projeto. Estão fora do escopo dela as ferramentas necessárias, insumos, hora técnica, gastos de energia elétrica e logístico com transportes.

TABELA 1 – CUSTOS GERAIS PARA FABRICAÇÃO

Item	Componente	Quantidade	Custo unitário (R\$)	Custo total (R\$)
1	Borne	1	1,50	1,50
2	Capacitor	1	0,20	0,20
3	CI LM555	1	1,50	1,50
4	CI CD4017	1	1,50	1,50
5	Conector bateria	1	2,50	2,50
6	Diodo	1	0,15	0,15
7	LED	1	0,25	0,25
8	Microfone	1	1,50	1,50
9	Bateria	1	5,00	5,00
10	Placa Fenolite	1	5,00	5,00
11	Relé	1	4,00	4,00
12	Resistor 1 k Ω	4	0,15	0,60
13	Resistor 10 k Ω	1	0,15	0,15
14	Resistor 100 k Ω	1	0,15	0,15
15	Potenciômetro 100 k Ω	1	1,70	1,70
16	Soquete 8 pinos	1	0,30	0,30
17	Soquete 16 pinos	1	0,40	0,40
18	Transistor	2	0,30	0,60
19	Fabricação (UP)	1	10,00	10,00
TOTAL:		23	36,25	37,00

FONTE: OS AUTORES

3 CONSIDERAÇÕES FINAIS

Inicialmente, pesquisamos e escolhemos o circuito, nos reunimos e montamos ele na matriz de contato aplicando os conceitos adquiridos no curso até o momento de forma prática e, conseqüentemente, verificando a funcionalidade do nosso projeto. Realizamos duas montagens do mesmo circuito em *protoboards* diferentes. O que havíamos escolhido tinha como finalidade, ao detectar uma frequência sonora próxima ao microfone, acender a lâmpada ligada através do relé e apagá-la segundos depois. O circuito também tinha um LED que indicava o funcionamento sem a necessidade de conectar a lâmpada. Quando realizamos os testes, o LED acionava e, após alguns instantes de tempo (variação devida a capacitância do capacitor utilizado), se apagava. Embora o circuito seja funcional, concluímos que não tinha tanto apelo comercial para uma fabricação em massa, porque entendemos que os futuros compradores teriam interesse em também desligar a lâmpada da mesma forma como a ligaram.

Por isso, resolvemos procurar outro circuito que o usuário pudesse controlar o desligamento da lâmpada, no momento que desejasse, não dependendo do capacitor. Encontramos um novo circuito, o qual atendia nosso objetivo, com bastante similaridade ao primeiro e a adição principal de um novo componente, o circuito integrado CD4017.

Com os componentes do segundo circuito, executamos os testes montando na *protoboard* e obtivemos o resultado esperado. Então, partimos para a elaboração do *layout* da PCI. Como aderimos a indicação do professor Jackson Milano para a fabricação, entramos em contato com a Universidade Positivo. Conforme solicitado, fizemos o *layout* no *software EasyEDA* e atendemos aos requisitos técnicos para a fabricação. Fazer o roteamento das trilhas, otimizando cada espaço e sem cruzar as linhas por causa da placa ser em uma única camada, se mostraram um grande desafio. Após várias tentativas, conseguimos aperfeiçoar o desenho e deixá-lo com apenas um *jumper* de conexão. Finalizamos gerando o arquivo *Gerber* e enviamos para a universidade. A produção levou apenas dois dias.

Após o recebimento da placa impressa, iniciamos a montagem dos componentes. A montagem exigiu muitos dos conceitos aprendidos na aula de Laboratório do primeiro semestre do curso de Engenharia Elétrica, matéria ministrada pelo professor Jackson Milano, onde foi necessário a realização dos processos de soldagem em circuitos impressos. Conseguimos fazer a montagem sem maiores problemas. Encaixamos os componentes e soldamos na placa. Ao fim, testamos novamente e alcançamos o resultado positivo na montagem.

Com poucos elementos já é possível iniciar projetos simples de eletrônica. Esse simples não significa fácil ou rápido de fazer, talvez não para a eletrônica. Exige tempo e dedicação. São muitas etapas a serem realizadas. Neste projeto tivemos a experiência de construir do zero um circuito. Essa experiência foi muito importante para o estudo destes acadêmicos de engenharia. A aprendizagem foi enorme. Houveram muitos momentos que comprovamos os conhecimentos e outros de busca de informação para passar o obstáculo. A nossa percepção é de que ainda temos muito caminho pela frente, muito o que estudar e aprender. Começamos a olhar diferente uma placa eletrônica, agora imaginando a forma como foi construída, a complexidade, o roteamento, as camadas, os testes, o desenvolvimento. Experimentos como esse ficam gravados no acervo pessoal do conhecimento e da experiência do aluno.

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FAIRCHILD. BC546, BC547, BC548, BC549, BC550 - NPN Epitaxial Silicon Transistor. 2014. Disponível em: <<https://www.mouser.com/datasheet/2/149/BC547-190204.pdf>>. Acesso em: 16 set. 2021.

ANEXO A – DATASHEET LM555

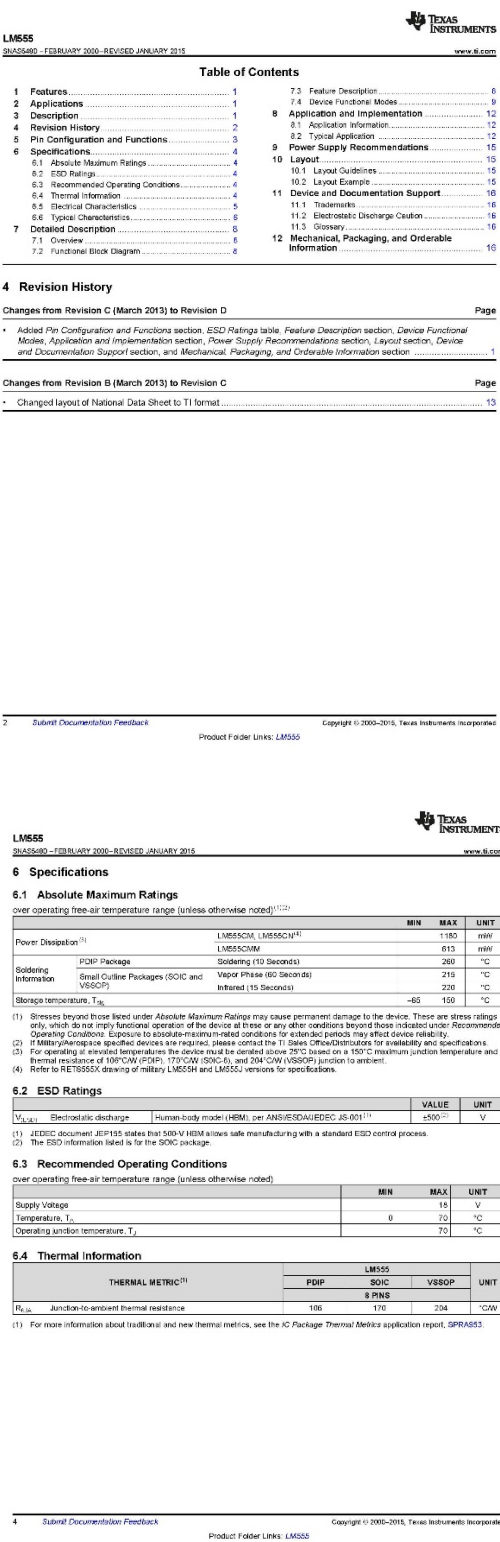
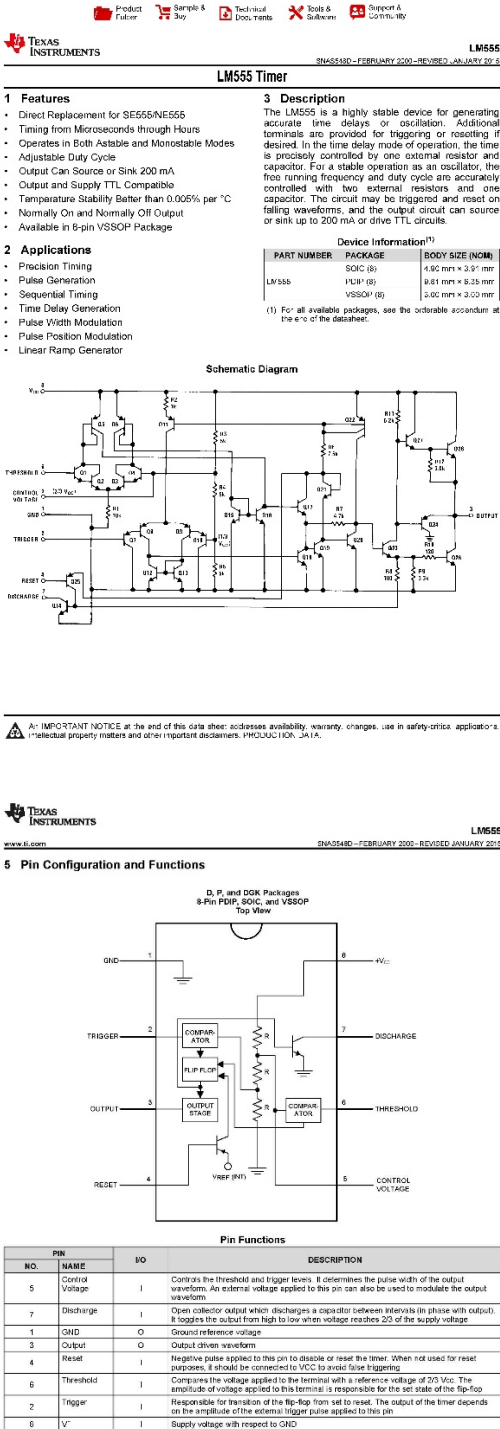


FIGURA 9 – DATASHEET LM555 – PARTE 1
FONTE: TEXAS INSTRUMENTS, 2015

		LM555	
www.ti.com		SNAS480 - FEBRUARY 2006 - REVISED JANUARY 2015	
6.5 Electrical Characteristics			
(T _A = 25 °C, V _{CC} = 5 V to 15 V, unless otherwise specified) ⁽¹⁾⁽²⁾			
PARAMETER	TEST CONDITIONS	MIN	TYP MAX UNIT
Supply Voltage		4.5	16 V
Supply Current	V _{CC} = 5 V, R _L = ∞	3	6 mA
	V _{CC} = 15 V, R _L = ∞	10	15 mA
	(Low State) ⁽³⁾		
Timing Error: Monostable			1%
Initial Accuracy			50
Drift with Temperature	R ₁ , R ₂ = 1 k to 100 kΩ C = 0.1 μF, °C		ppm/°C
Accuracy over Temperature			1.5%
Drift with Supply			0.1%
Timing Error: Astable			
Initial Accuracy			2.25
Drift with Temperature	R ₁ , R ₂ = 1 k to 100 kΩ C = 0.1 μF, °C		ppm/°C
Accuracy over Temperature			3.0%
Drift with Supply			0.39%
Threshold Voltage	V _{CC} = 15 V	0.687	π V _{CC}
Trigger Voltage	V _{CC} = 5 V	5	V
	V _{CC} = 5 V	1.87	V
Trigger Current		0.5	0.9 μA
Reset Voltage		0.4	0.5 V
Reset Current		0.1	0.6 mA
Threshold Current		0.1	0.26 μA
Control Voltage Level	V _{CC} = 15 V	9	10 11 V
	V _{CC} = 5 V	2.6	3.33 4 V
Pin 7 Leakage Output High		1	100 nA
Pin 7 Sat ⁽⁴⁾			
Output Low	V _{CC} = 15 V, I _L = 15 mA	180	mV
Output Low	V _{CC} = 4.5 V, I _L = 4.5 mA	90	200 mV
Output Voltage Drop (Low)	V _{CC} = 15 V		
	I _{OUT} = 10 mA	0.1	0.25 V
	I _{OUT} = 50 mA	0.4	0.75 V
	I _{OUT} = 100 mA	2	2.5 V
	I _{OUT} = 200 mA		2.5 V
	V _{CC} = 5 V		
	I _{OUT} = 8 mA		V
	I _{OUT} = 0.5 mA		V
	I _{OUT} = 0.25 mA	0.25	0.35 V

(1) All voltages are measured with respect to the ground pin, unless otherwise specified.
(2) Absolute Maximum Ratings indicate limits beyond which damage to the device may occur. Recommended Operating Conditions indicate conditions for which the device is functional, but do not ensure specific performance limits. Electrical Characteristics state DC and AC electrical specifications under particular test conditions which ensures specific performance limits. This assumes that the device is within the Recommended Operating Conditions. Specifications are not ensured for parameters where no limit is given; however, the typical value is a good indication of device performance.
(3) Supply current when output high typically 1 mA less at V_{CC} = 5 V.
(4) Tested at V_{CC} = 5 V and V_{OUT} = 15 V.
(5) This will determine the maximum value of R₁ + R₂ for 15 V operation. The maximum total (R₁ + R₂) is 20 MΩ.
(6) No protection against excessive pin 7 current is necessary providing the package dissipation rating will not be exceeded.

		LM555	
www.ti.com		SNAS480 - FEBRUARY 2006 - REVISED JANUARY 2015	

Electrical Characteristics (continued)				
(T _A = 25 °C, V _{CC} = 5 V to 15 V, unless otherwise specified) ⁽¹⁾⁽²⁾				
PARAMETER	TEST CONDITIONS	MIN	TYP	MAX. UNIT
Output: Voltage Drop (High)	I _{source} = 200 mA, V _{CC} = 15 V		12.5	V
	I _{source} = 100 mA, V _{CC} = 15 V	12.75	12.5	V
	V _{CC} = 5 V	2.75	3.3	V
Rise Time of Output			100	ns
Fall Time of Output			100	ns

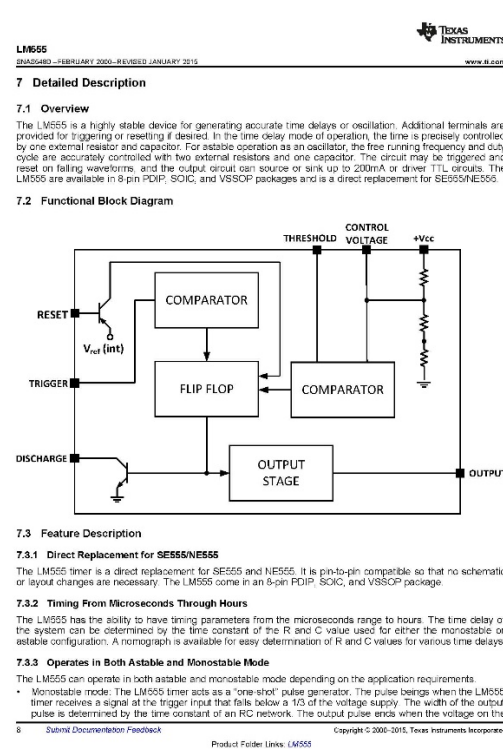
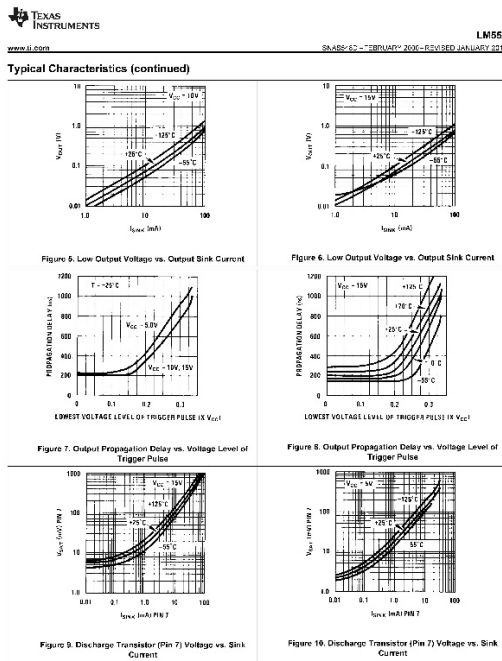
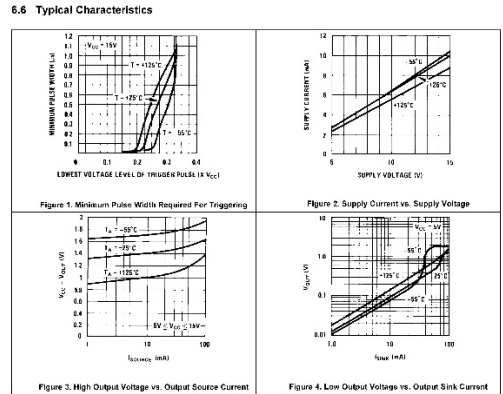


FIGURA 10 – DATASHEET LM555 – PARTE 2
FONTE: TEXAS INSTRUMENTS, 2015



www.ti.com

LM555

Feature Description (continued)

capacitor equals 2/5 of the supply voltage. The output pulse width can be extended or shortened depending on the application by adjusting the R and C values.

- **Astable (free-running) mode:** The LM555 timer can operate as an oscillator and puts out a continuous stream of rectangular pulses having a specified frequency. The frequency of the pulse stream depends on the values of R_A , R_B , and C.

7.4 Device Functional Modes

7.4.1 Monostable Operation

In this mode of operation, the timer functions as a one-shot (Figure 11). The external capacitor is initially held discharged by a transistor inside the timer. Upon application of a negative trigger pulse of less than $1/3 V_{CC}$ to pin 2, the flip-flop is set which both releases the short circuit across the capacitor and drives the output high.

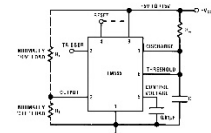


Figure 11. Monostable

The voltage across the capacitor then increases exponentially for a period of $t = 1.1 R_A C$, at the end of which time the voltage equals $2/5 V_{CC}$. The comparator then resets the flip-flop which in turn discharges the capacitor and drives the output to its low state. Figure 12 shows the waveforms generated in this mode of operation. Since the charge and the threshold level of the comparator are both directly proportional to supply voltage, the timing interval is independent of supply.



$V_{CC} = 5V$
 TIME = 0.1 ms/DIV.
 $R_A = 9.1k\Omega$
 $C = 0.01\mu F$

Top Trace: Output 5V/DIV.
 Middle Trace: Output 5V/DIV.
 Bottom Trace: Capacitor Voltage 2V/DIV.

Figure 12. Monostable Waveforms

During the timing cycle when the output is high, the further application of a trigger pulse will not effect the circuit so long as the trigger input is returned high at least 10 μs before the end of the timing interval. However the circuit can be reset during this time by the application of a negative pulse to the reset terminal (pin 4). The output will then remain in the low state until a trigger pulse is again applied.

When the reset function is not in use, TI recommends connecting the Reset pin to V_{CC} to avoid any possibility of false triggering.

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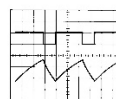
Product Folder Link: LM555



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LM555

Device Functional Modes (continued)



$V_{CC} = 5V$
 TIME = 0.1 ms/DIV.
 $R_A = 3.3k\Omega$
 $R_B = 3.3k\Omega$
 $C = 0.01\mu F$

Top Trace: Output 5V/DIV.
 Bottom Trace: Capacitor Voltage 2V/DIV.

Figure 13. Astable Waveforms

The charge time (output high) is given by:

$$t_1 = 0.693 (R_A + R_B) C$$

And the discharge time (output low) by:

$$t_2 = 0.693 R_B C$$

Thus the total period is:

$$T = t_1 + t_2 = 0.693 (R_A + 2R_B) C$$

The frequency of oscillation is:

$$f = \frac{1}{T} = \frac{1.44}{0.693 (R_A + 2R_B) C}$$

Figure 16 may be used for quick determination of these RC values.

The duty cycle is:

$$D = \frac{t_1}{T} = \frac{R_A + R_B}{R_A + 2R_B}$$

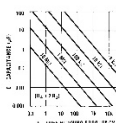


Figure 16. Free Running Frequency

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Device Functional Modes (continued)

Figure 13 is a nomograph for easy determination of R, C values for various time delays.

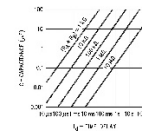


Figure 13. Time Delay

7.4.2 Astable Operation

If the circuit is connected as shown in Figure 14 (pins 2 and 6 connected) it will trigger itself and free run as a multivibrator. The external capacitor charges through $R_A + R_B$ and discharges through R_B . Thus the duty cycle may be precisely set by the ratio of these two resistors.

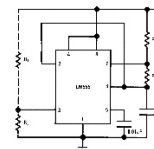


Figure 14. Astable

In this mode of operation, the capacitor charges and discharges between $1/3 V_{CC}$ and $2/3 V_{CC}$. As in the triggered mode, the charge and discharge times, and therefore the frequency are independent of the supply voltage.

Figure 15 shows the waveforms generated in this mode of operation.

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8 Application and Implementation

NOTE

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes. Customers should validate and test their design implementation to confirm system functionality.

8.1 Application Information

The LM555 timer can be used a various configurations, but the most commonly used configuration is in monostable mode. A typical application for the LM555 timer in monostable mode is to turn on an LED for a specific time duration. A pushbutton is used as the trigger to output a high pulse when trigger pin is pulsed low. This simple application can be modified to fit any application requirement.

8.2 Typical Application

Figure 17 shows the schematic of the LM555 that flashes an LED in monostable mode.

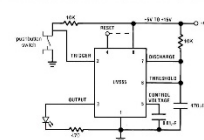


Figure 17. Schematic of Monostable Mode to Flash an LED

8.2.1 Design Requirements

The main design requirement for this application requires calculating the duration of time for which the output stays high. The duration of time is dependent on the R and C values (as shown in Figure 17) and can be calculated by:

$$t = 1.1 \times R \times C \text{ seconds}$$

(6)

8.2.2 Detailed Design Procedure

To allow the LED to flash on for a noticeable amount of time, a 5 second time delay was chosen for this application. By using Equation 6, RC equals 4.54s. If R is selected as 100 k Ω , C = 45.4 μF . The values of R = 100 k Ω and C = 47 μF was selected based on standard values of resistors and capacitors. A momentary push button switch connected to ground is connected to the trigger input with a 10-k current limiting resistor pullup to the supply voltage. When the push button is pressed, the trigger pin goes to GND. An LED is connected to the output pin with a current limiting resistor in series from the output of the LM555 to GND. The reset pin is not used and was connected to the supply voltage.

8.2.2.1 Frequency Divider

The monostable circuit of Figure 11 can be used as a frequency divider by adjusting the length of the timing cycle. Figure 18 shows the waveforms generated in a divide by three circuit.

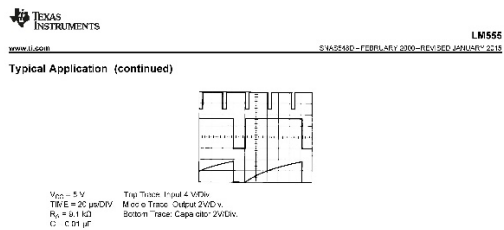


Figure 18. Frequency Divider

5.2.2.2 Additional Information

Lower comparator storage time can be as long as 10 μ s when pin 2 is driven fully to ground for triggering. This limits the monostable pulse width to 10 μ s minimum.
Delay time reset to output is 0.47 μ s typical. Minimum reset pulse width must be 0.3 μ s, typical.
Pin 7 current switches within 30 ns of the output (pin 3) voltage.

5.2.3 Application Curves

The data shown below was collected with the circuit used in the typical applications section. The LM555 was configured in the monostable mode with a time delay of 5.17 s. The waveforms correspond to:

- Top Waveform (Yellow) – Capacitor voltage
- Middle Waveform (Green) – Trigger
- Bottom Waveform (Purple) – Output

As the trigger pin pulses low, the capacitor voltage starts charging and the output goes high. The output goes low as soon as the capacitor voltage reaches 2/3 of the supply voltage, which is the time delay set by the R and C value. For this example, the time delay is 5.17 s.



Figure 19. Trigger, Capacitor Voltage, and Output Waveforms in Monostable Mode

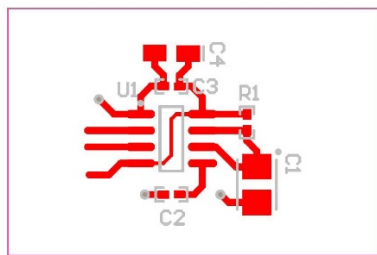
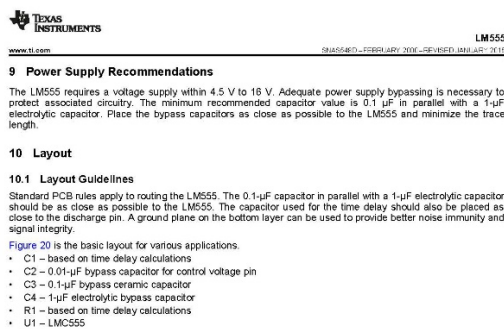


Figure 20. Layout Example

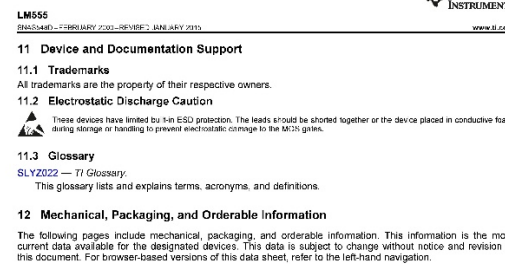


FIGURA 12 – DATASHEET LM555 – PARTE 4
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PACKAGING INFORMATION

Orderable Device	Status (1)	Package Type	Package Drawing	Pins	Package Qty	Eco Plan (2)	Lead finish / Ball material (3)	MSL Peak Temp (4)	Op Temp (°C)	Device Marking (4a)	Sample #
LM555CM	NRND	SOIC	D	8	95	Non-RoHS & Green	Call TI	Call TI	0 to 70	LM555CM	
LM555CM/NOPB	ACTIVE	SOIC	D	8	95	RoHS & Green	SN	Level-1-260C-UNLIM	0 to 70	LM555CM	Samples
LM555CMM	NRND	VSSOP	DGK	8	1000	Non-RoHS & Green	Call TI	Call TI	0 to 70	Z95	
LM555CMM/NOPB	ACTIVE	VSSOP	DGK	8	1000	RoHS & Green	SN	Level-1-260C-UNLIM	0 to 70	Z95	Samples
LM555CMMX/NOPB	ACTIVE	VSSOP	DGK	8	3500	RoHS & Green	SN	Level-1-260C-UNLIM	0 to 70	Z95	Samples
LM555CMX	NRND	SOIC	D	8	2500	Non-RoHS & Green	Call TI	Call TI	0 to 70	LM555CM	
LM555CMX/NOPB	ACTIVE	SOIC	D	8	2500	RoHS & Green	SN	Level-1-260C-UNLIM	0 to 70	LM555CM	Samples
LM555CN/NOPB	ACTIVE	PDIP	P	8	40	RoHS & Green	Call TI SN	Level-1-NA-UNLIM	0 to 70	LM555CN	Samples

(1) The marketing status values are defined as follows:

ACTIVE: Product device recommended for new designs.

LIFEBUY: TI has announced that the device will be discontinued, and a life-buy period is in effect.

NRND: Not recommended for new designs. Device is in production to support existing customers, but TI does not recommend using this part in a new design.

PREVIEW: Device has been announced but is not in production. Samples may or may not be available.

OBsolete: TI has discontinued the production of the device.

(2) RoHS: TI defines "RoHS" to mean semiconductor products that are compliant with the current EU RoHS requirements for all 10 RoHS substances, including the requirement that RoHS substances do not exceed 0.1% by weight in homogeneous materials. Where designed to be soldered at high temperatures, "RoHS" products are suitable for use in specified lead-free processes. TI may not use these types of products as "Pb-Free".

RoHS Exempt: TI defines "RoHS Exempt" to mean products that contain lead but are compliant with EU RoHS pursuant to a specific EU RoHS exemption.

Green: TI defines "Green" to mean the product conforms to JEDEC low halogen requirements of ≤1000ppm threshold. Any other device-based standards must also meet the ≤1000ppm threshold requirement.

(3) MSL, Peak Temp. - The Moisture Sensitivity Level rating according to the JEDEC industry standard classifications, and peak solder temperature.

(4) There may be additional marking, which relates to the logo, lot trace code information, or the full part number category on the device.

Addendum-Page 1

(4a) Multiple Device Markings will be inside parentheses. Only one Device Marking contained in parentheses and separated by a "~" will appear on a device. If a line is indicated then it is a continuation of the previous line and the two combined represent the entire Device Marking for that device.

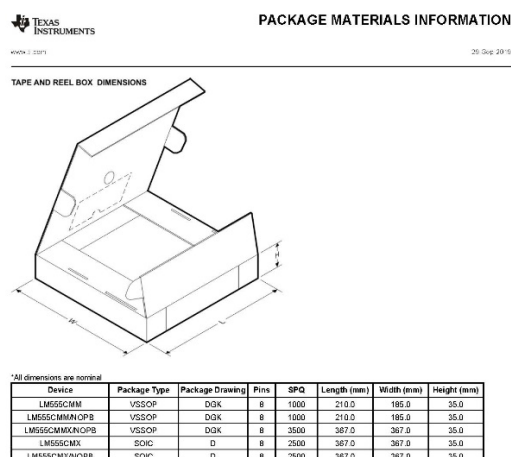
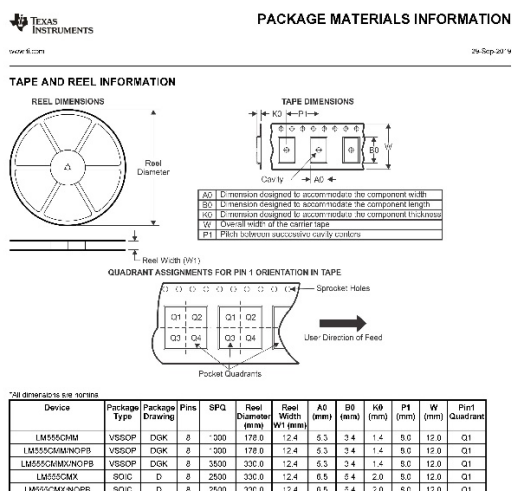
(5) Lead finish/Ball material - Orderable Devices may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

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Addendum-Page 2

FIGURA 13 – DATASHEET LM555 – PARTE 5
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Pack Materials-Page 2

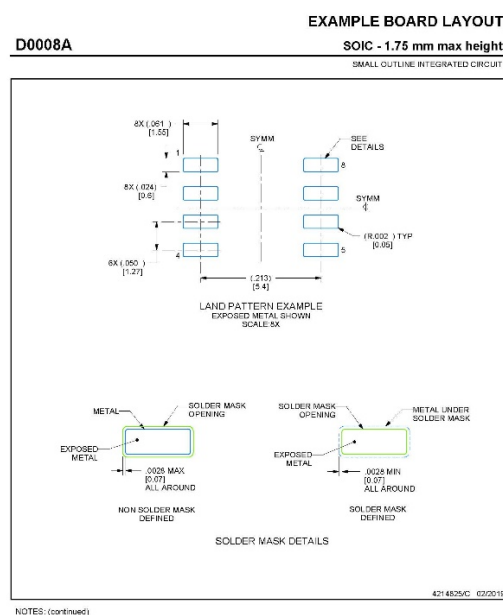
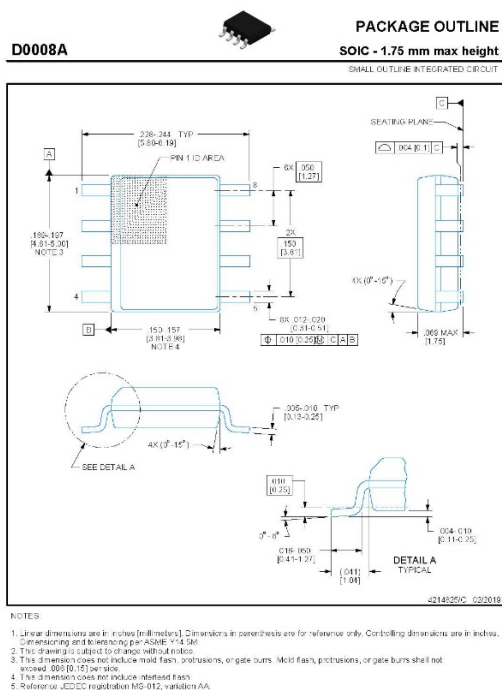
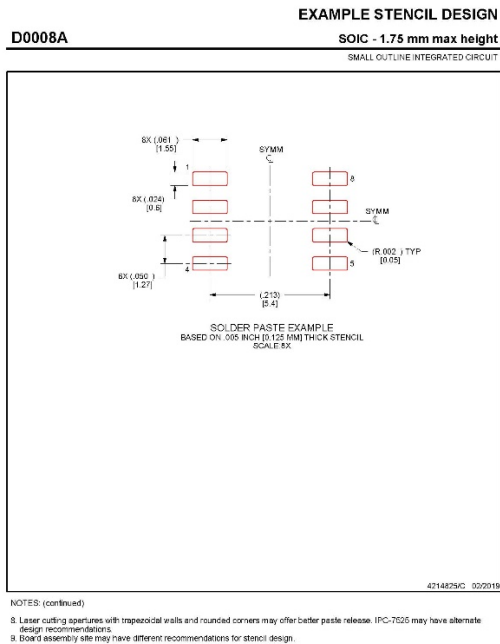
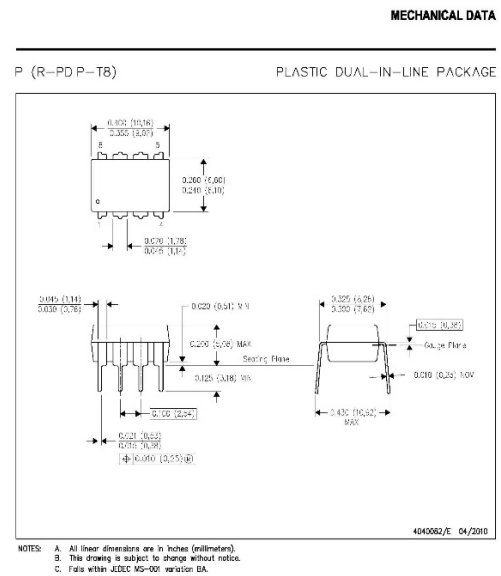
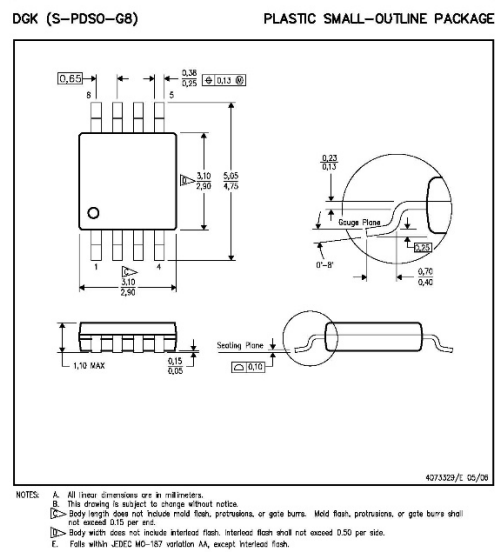


FIGURA 14 – DATASHEET LM555 – PARTE 6
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MECHANICAL DATA



LAND PATTERN DATA

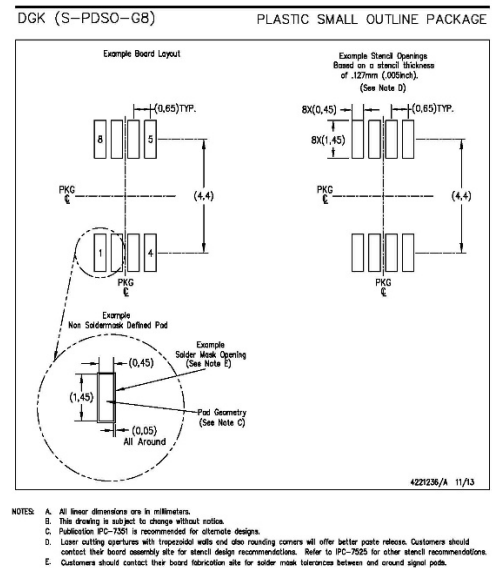


FIGURA 15 – DATASHEET LM555 – PARTE 7
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FIGURA 16 – DATASHEET LM555 – PARTE 8
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Da Figura 9 a Figura 16, com tamanho reduzido em virtude da grandeza de informações e por se tratar de uma referência neste anexo, reunimos todas as 27 páginas do *datasheet* disponibilizado pelo fabricante em oito partes.

ANEXO B – DATASHEET CD4017



CD4017B, CD4022B Types

CMOS Counter/Dividers

High-Voltage Types (20-Volt Rating)

CD4017B—Decade Counter with 10 Decoded Outputs

CD4022B—Oscillator Counter with 8 Decoded Outputs

Features:

- Fully static operation
- Medium-speed operation: 10 MHz typ. at $V_{DD} = 10\text{ V}$
- Standardized, symmetrical output characteristics
- 100% tested for quiescent current at 20 V
- 5-V, 10-V, and 15-V parametric ratings
- Meets all requirements of JEDEC Test Method Standard No. 13A, "Standard Specifications for Description of 'B' Series CMOS Devices"

Applications:

- Decade counter/decimal decade display (CD4017B)
- Binary counter/decoder
- Frequency divider
- Counter control/limiter
- Divide-by-N counting
- For further application information, see ICAN-515B "COSMOS MSI Counter and Register Design and Applications"

The CD4017B and CD4022B types are supplied in 16-lead ceramic dual-in-line, ceramic packages (73A suffix), 16-lead small-outline plastic package (E suffix), 16-lead small-outline package (NSI suffix), and 16-lead thin-shrink small-outline package (PWH and PWS suffixes). The CD4017B type also is available in 16-lead small-outline package (JF and JBS suffixes).

These counters are advanced one count at the positive clock signal transition if the CLOCK (INHIBIT) input is low. Counter advancement via the clock line is inhibited when the CLOCK (INHIBIT) signal is high. A high RESET signal starts the counter to its zero count. Use of the Johnson counter configuration permits high-speed operation. 2-input decode gating and spike-free decoded outputs are normally low and go high only at their respective decoded time slot. Each decoded output remains high for one full clock cycle. A CARRY-OUT signal appears one cycle every 10 clock input cycles in the CD4017B or every 8 clock input cycles in the CD4022B and is used to cascade the succeeding device in a multi-device counting chain.

RECOMMENDED OPERATING CONDITIONS

For maximum reliability, normal operating conditions should be selected so that operation is always within the following ranges:

CHARACTERISTICS

Supply Voltage Range (For $T_A = \text{Full Package Temperature Range}$)

Clock Input Frequency, f_{CL}

Clock Pulse Width, t_{WP}

Clock Rise & Fall Time, t_{CL} , t_{FL}

Clock Inhibit Setup Time, t_{SI}

Reset Pulse Width, t_{RPW}

Reset Removal Time, t_{RR}

Only if t_{SI} is used as the clock input, if Pin 13 is used as the clock input and Pin 14 is tied high for addressing count on negative transition of the clock, rise and fall time should be $\leq 15\text{ ns}$.

CD4017B, CD4022B Types

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Clock Rise & Fall Time, t_{CL} , t_{FL}

Clock Inhibit Setup Time, t_{SI}

Reset Pulse Width, t_{RPW}

Reset Removal Time, t_{RR}

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CD4017B, CD4022B Types

CD4017B, CD4022B Types

CD4017B, CD4022B Types

CD4017B, CD4022B Types

CD4017B, CD4022B Types

CD4017B, CD4022B Types

CD4017B, CD4022B Types

CD4017B, CD4022B Types

CD4017B, CD4022B Types

CD4017B, CD4022B Types

CD4017B, CD4022B Types

CD4017B, CD4022B Types

CD4017B, CD4022B Types

CD4017B, CD4022B Types

CD4017B, CD4022B Types

CD4017B, CD4022B Types

CD4017B, CD4022B Types

CD4017B, CD4022B Types

CD4017B, CD4022B Types

CD4017B, CD4022B Types

CD4017B, CD4022B Types

CD4017B, CD4022B Types

CD4017B, CD4022B Types

CD4017B, CD4022B Types

CMOS Counter/Dividers

High-Voltage Types (20-Volt Rating)

CD4017B—Decade Counter with 10 Decoded Outputs

CD4022B—Oscillator Counter with 8 Decoded Outputs

Features:

- Fully static operation
- Medium-speed operation: 10 MHz typ. at $V_{DD} = 10\text{ V}$
- Standardized, symmetrical output characteristics
- 100% tested for quiescent current at 20 V
- 5-V, 10-V, and 15-V parametric ratings
- Meets all requirements of JEDEC Test Method Standard No. 13A, "Standard Specifications for Description of 'B' Series CMOS Devices"

Applications:

- Decade counter/decimal decade display (CD4017B)
- Binary counter/decoder
- Frequency divider
- Counter control/limiter
- Divide-by-N counting
- For further application information, see ICAN-515B "COSMOS MSI Counter and Register Design and Applications"

The CD4017B and CD4022B types are supplied in 16-lead ceramic dual-in-line, ceramic packages (73A suffix), 16-lead small-outline plastic package (E suffix), 16-lead small-outline package (NSI suffix), and 16-lead thin-shrink small-outline package (PWH and PWS suffixes). The CD4017B type also is available in 16-lead small-outline package (JF and JBS suffixes).

These counters are advanced one count at the positive clock signal transition if the CLOCK (INHIBIT) input is low. Counter advancement via the clock line is inhibited when the CLOCK (INHIBIT) signal is high. A high RESET signal starts the counter to its zero count. Use of the Johnson counter configuration permits high-speed operation. 2-input decode gating and spike-free decoded outputs are normally low and go high only at their respective decoded time slot. Each decoded output remains high for one full clock cycle. A CARRY-OUT signal appears one cycle every 10 clock input cycles in the CD4017B or every 8 clock input cycles in the CD4022B and is used to cascade the succeeding device in a multi-device counting chain.

RECOMMENDED OPERATING CONDITIONS

For maximum reliability, normal operating conditions should be selected so that operation is always within the following ranges:

CHARACTERISTICS

Supply Voltage Range (For $T_A = \text{Full Package Temperature Range}$)

Clock Input Frequency, f_{CL}

CD4017B, CD4022B Types

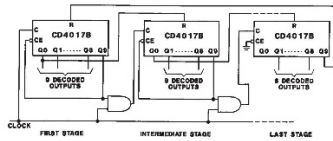
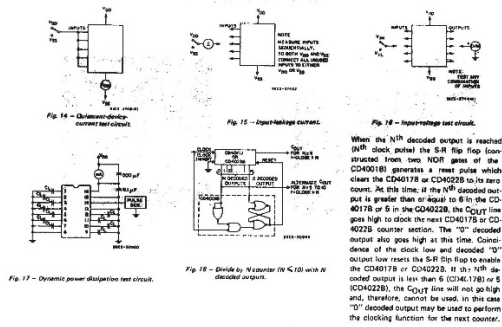
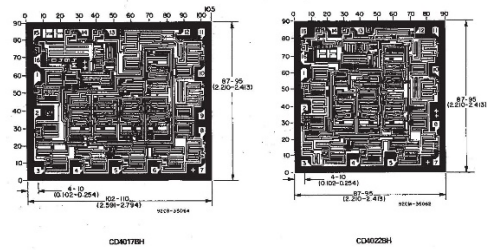


Fig. 18 - Cascading the CD4017B.

3-55

CD4017B, CD4022B Types

CHIP DIMENSIONS AND PAD LAYOUTS



Dimensions in parentheses are in millimeters and are derived from the basic inch dimensions as indicated. Grid graduations are in mils (10⁻³ inch).

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PACKAGE OPTION ADDENDUM

14-Aug-2021

PACKAGING INFORMATION

Orderable Device	Status (1)	Package Type	Package Drawing	Pins	Package Qty	Eco Plan (2)	Lead finish/Ball material (6)	MSL Peak Temp (3)	Op Temp (°C)	Device Marking (4/5)	Samples
CD4017BE	ACTIVE	PDIP	N	16	25	RoHS & Green	NIPDAU	N / A for Pkg Type	-55 to 125	CD4017BE	Samples
CD4017BEE4	ACTIVE	PDIP	N	16	25	RoHS & Green	NIPDAU	N / A for Pkg Type	-55 to 125	CD4017BE	Samples
CD4017BF	ACTIVE	CDIP	J	16	1	Non-RoHS & Green	SNPB	N / A for Pkg Type	-55 to 125	CD4017BF	Samples
CD4017BF3A	ACTIVE	CDIP	J	16	1	Non-RoHS & Green	SNPB	N / A for Pkg Type	-55 to 125	CD4017BF3A	Samples
CD4017BM	ACTIVE	SOIC	D	16	40	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-55 to 125	CD4017BM	Samples
CD4017BM96	ACTIVE	SOIC	D	16	2500	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-55 to 125	CD4017BM	Samples
CD4017BM96G4	ACTIVE	SOIC	D	16	2500	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-55 to 125	CD4017BM	Samples
CD4017BMG4	ACTIVE	SOIC	D	16	40	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-55 to 125	CD4017BM	Samples
CD4017BNSR	ACTIVE	SO	NS	16	2000	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-55 to 125	CD4017B	Samples
CD4017BPW	ACTIVE	TSSOP	PW	16	90	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-55 to 125	CM017B	Samples
CD4017BPWR	ACTIVE	TSSOP	PW	16	2000	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-55 to 125	CM017B	Samples
CD4017BPWRE4	ACTIVE	TSSOP	PW	16	2000	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-55 to 125	CM017B	Samples
CD4022BE	ACTIVE	PDIP	N	16	25	RoHS & Green	NIPDAU	N / A for Pkg Type	-55 to 125	CD4022BE	Samples
CD4022BEE4	ACTIVE	PDIP	N	16	25	RoHS & Green	NIPDAU	N / A for Pkg Type	-55 to 125	CD4022BE	Samples
CD4022BF	ACTIVE	CDIP	J	16	1	Non-RoHS & Green	SNPB	N / A for Pkg Type	-55 to 125	CD4022BF	Samples
CD4022BF3A	ACTIVE	CDIP	J	16	1	Non-RoHS & Green	SNPB	N / A for Pkg Type	-55 to 125	CD4022BF3A	Samples
CD4022BNSR	ACTIVE	SO	NS	16	2000	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-55 to 125	CD4022B	Samples
CD4022BNSRE4	ACTIVE	SO	NS	16	2000	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-55 to 125	CD4022B	Samples
CD4022BPW	ACTIVE	TSSOP	PW	16	90	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-55 to 125	CM022B	Samples

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FIGURA 18 – DATASHEET CD4017 – PARTE 2
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PACKAGE OPTION ADDENDUM

14-Aug-2021

Orderable Device	Status (1)	Package Type	Package Drawing	Pins	Package Qty	Eco Plan (2)	Lead finish/ Ball material (6)	MSL Peak Temp (3)	Op Temp (°C)	Device Marking (4,5)	Samples
CD4022BPWR	ACTIVE	TSSOP	PW	16	2000	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-55 to 125	CM022B	Samples
CD4022BPWRG4	ACTIVE	TSSOP	PW	16	2000	RoHS & Green	NIPDAU	Level-1-260C-UNLIM	-55 to 125	CM022B	Samples
JM38510/05651BEA	ACTIVE	CDIP	J	16	1	Non-RoHS & Green	SNPB	N / A for Pkg Type	-55 to 125	JM38510/ 05651BEA	Samples
M38510/05651BEA	ACTIVE	CDIP	J	16	1	Non-RoHS & Green	SNPB	N / A for Pkg Type	-55 to 125	JM38510/ 05651BEA	Samples

(1) The marketing status values are defined as follows:

ACTIVE: Product device recommended for new designs.**LIFEBUY:** TI has announced that the device will be discontinued, and a lifetime-buy period is in effect.**NRND:** Not recommended for new designs. Device is in production to support existing customers, but TI does not recommend using this part in a new design.**PREVIEW:** Device has been announced but is not in production. Samples may or may not be available.**OBSOLETE:** TI has discontinued the production of the device.(2) **RoHS:** TI defines "RoHS" to mean semiconductor products that are compliant with the current EU RoHS requirements for all 10 RoHS substances, including the requirement that RoHS substance do not exceed 0.1% by weight in homogeneous materials. Where designed to be soldered at high temperatures, "RoHS" products are suitable for use in specified lead-free processes. TI may reference these types of products as "Pb-Free".**RoHS Exempt:** TI defines "RoHS Exempt" to mean products that contain lead but are compliant with EU RoHS pursuant to a specific EU RoHS exemption.**Green:** TI defines "Green" to mean the content of Chlorine (Cl) and Bromine (Br) based flame retardants meet JS709B low halogen requirements of <=1000ppm threshold. Antimony trioxide based flame retardants must also meet the <=1000ppm threshold requirement.(3) **MSL, Peak Temp.:** - The Moisture Sensitivity Level rating according to the JEDEC industry standard classifications, and peak solder temperature.

(4) There may be additional marking, which relates to the logo, the lot trace code information, or the environmental category on the device.

(5) Multiple Device Markings will be inside parentheses. Only one Device Marking contained in parentheses and separated by a "~" will appear on a device. If a line is indented then it is a continuation of the previous line and the two combined represent the entire Device Marking for that device.

(6) **Lead finish/Ball material** - Orderable Devices may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.**Important Information and Disclaimer:** The information provided on this page represents TI's knowledge and belief as of the date that it is provided. TI bases its knowledge and belief on information provided by third parties, and makes no representation or warranty as to the accuracy of such information. Efforts are underway to better integrate information from third parties. TI has taken and continues to take reasonable steps to provide representative and accurate information but may not have conducted destructive testing or chemical analysis on incoming materials and chemicals. TI and TI suppliers consider certain information to be proprietary, and thus CAS numbers and other limited information may not be available for release.

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PACKAGE OPTION ADDENDUM

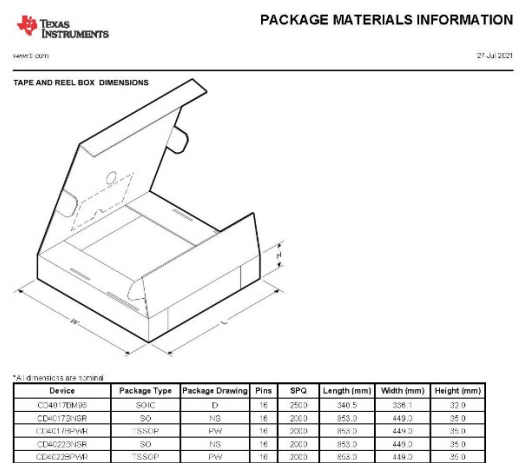
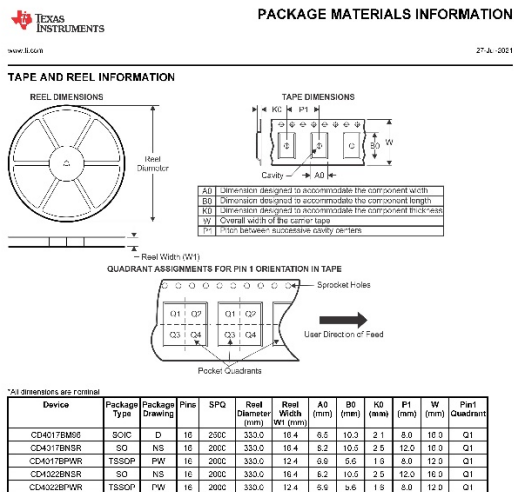
14-Aug-2021

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OTHER QUALIFIED VERSIONS OF CD4017B, CD4017B-MIL, CD4022B, CD4022B-MIL :• **Catalog :** CD4017B, CD4022B• **Military :** CD4017B-MIL, CD4022B-MIL**NOTE: Qualified Version Definitions:**• **Catalog** - TI's standard catalog product• **Military** - QML certified for Military and Defense Applications

Addendum-Page 3

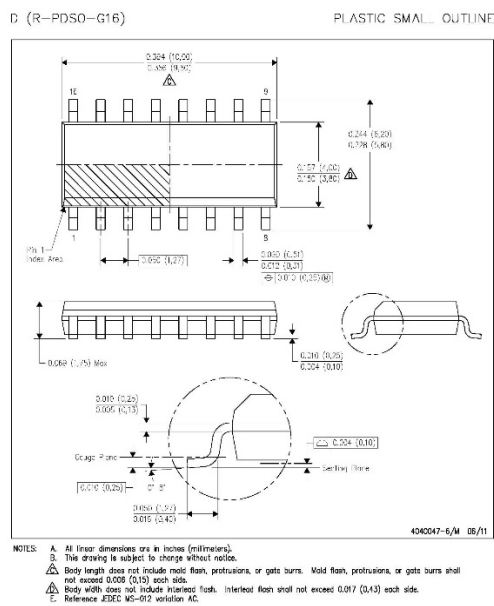
FIGURA 19 – DATASHEET CD4017 – PARTE 3
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Package Materials Page 1

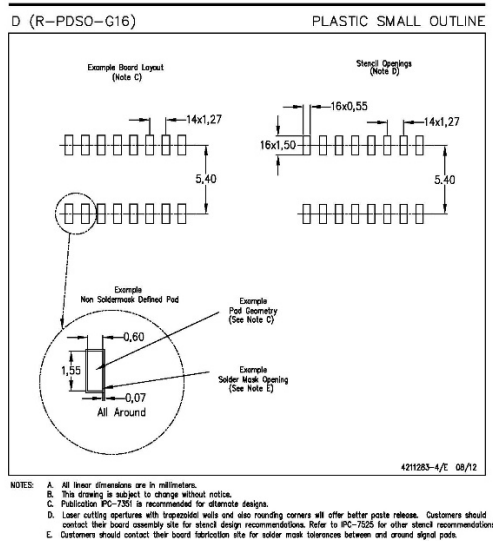
Package Materials Page 2

MECHANICAL DATA



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FIGURA 20 – DATASHEET CD4017 – PARTE 4
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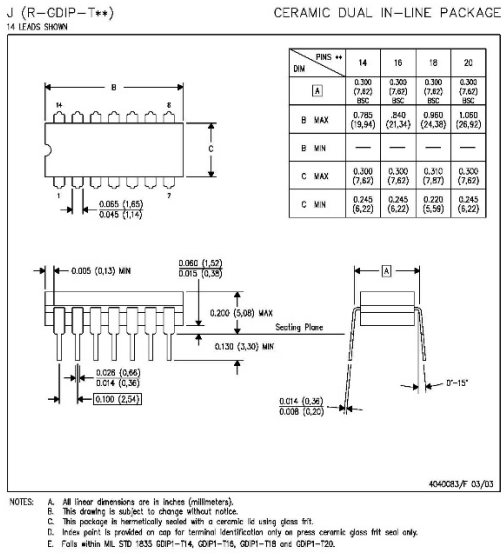
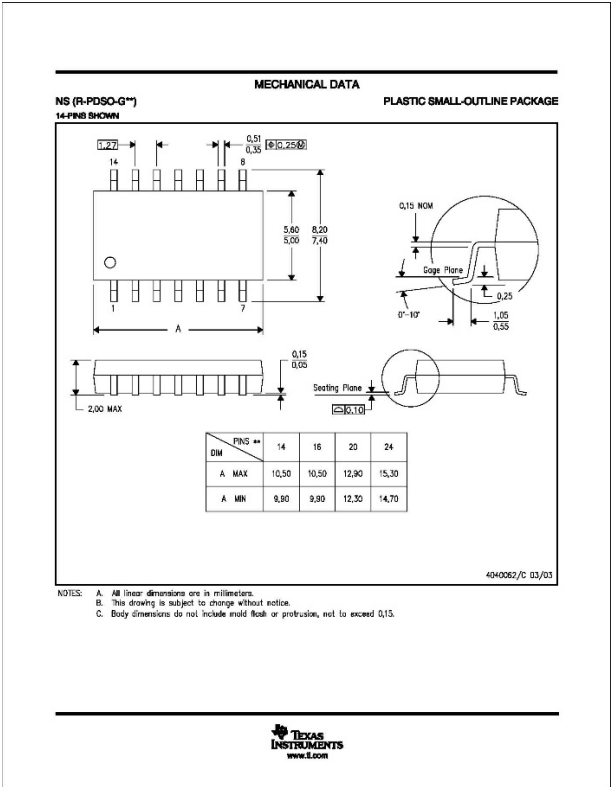
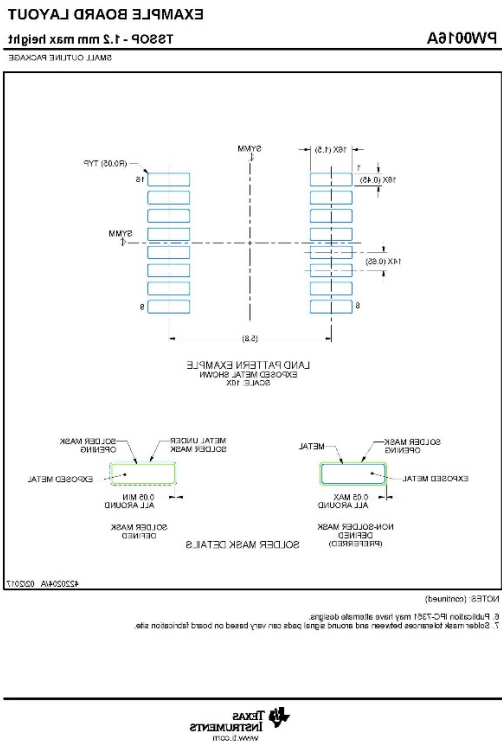
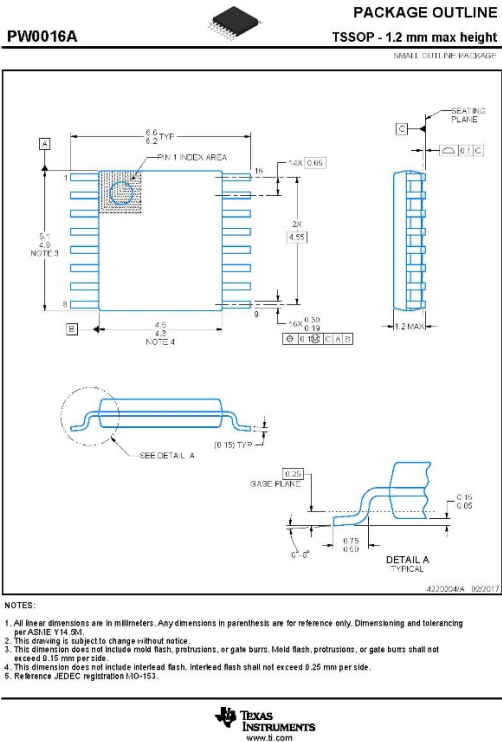
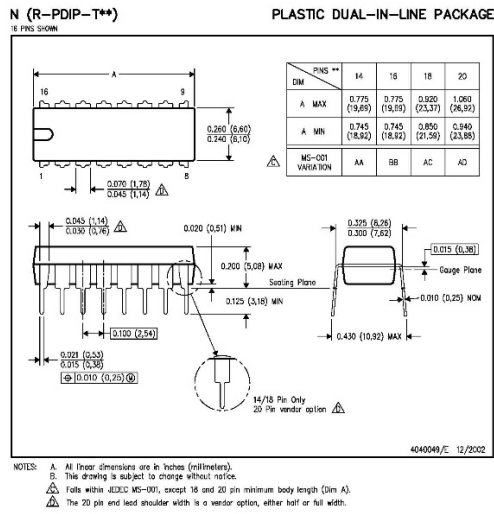


FIGURA 21 – DATASHEET CD4017 – PARTE 5
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MECHANICAL DATA



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FIGURA 22 – DATASHEET CD4017 – PARTE 6
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Reunimos em seis partes, Figura 17 a Figura 22, novamente em tamanho reduzido, as 20 páginas do *datasheet* do contador CD4017.